# Machine Learning — Lecture 8

## Unsupervised Learning and Dimensionality Reduction

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# Lecture 8 at a Glance

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# Unsupervised Learning and Dimensionality Reduction

## Course Learning Outcomes

**CLO4:** Apply supervised and **unsupervised learning** techniques to solve practical problems and interpret their outcomes.

**CLO2:** Analyze datasets and apply appropriate preprocessing, transformation, and feature engineering techniques.

## Lecture purpose

- Move from labeled prediction to discovering structure in unlabeled data.
- Understand clustering and K-Means.
- Perform manual K-Means assignment and update steps.
- Interpret clusters carefully.
- Understand dimensionality reduction and PCA.
- Explain how PCA transforms features while preserving major variation.

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💡 Main idea: unsupervised learning studies structure in $X$ when there is no target label $y$.

# Learning Objectives

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# By the end of Lecture 8, you will be able to:

1. Distinguish supervised and **unsupervised learning**.
2. Explain clustering as grouping observations without labels.
3. Describe the K-Means objective and algorithm.
4. Compute one K-Means assignment/update iteration manually.
5. Explain the effect of scaling, initialization, and the choice of $K$.
6. Interpret clusters as model outputs, not guaranteed real categories.
7. Describe hierarchical clustering conceptually.
8. Explain dimensionality reduction and why it is useful.
9. Describe PCA using variance, principal components, and projection.
10. Interpret explained variance and limitations of PCA.

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# Lecture Roadmap

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# From unlabeled data to interpretable structure

| Learning phase | Parts | Core question |
|---|---|---|
| **1. Understand unlabeled data and similarity** | **I–II** | What patterns can we investigate without a target, and what makes two examples similar? |
| **2. Discover clusters** | **III–IV** | How do K-Means and hierarchical clustering form groups, and what are their assumptions? |
| **3. Reduce dimensionality** | **V** | How does PCA represent many features using a few directions of variation? |
| **4. Validate and apply the results** | **VI–VII** | Are the discovered patterns stable, interpretable, and useful for decisions? |

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💡 **Practice & consolidation:** apply K-Means, interpret the choice of $K$, reason about PCA variance, and identify common unsupervised-learning mistakes.

# Part I

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# From Supervised to Unsupervised Learning

Most previous lectures assumed labeled data. Now we ask what can be learned when **labels** are absent.

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# Supervised vs. Unsupervised Learning

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# What changes?

The key difference is whether the training data include **known target values**.

| Aspect | Supervised learning | Unsupervised learning |
|---|---|---|
| **Training data** | Features and targets $(X, y)$ | Features only $X$ |
| **Goal** | Learn to predict $\hat{y}$ from $X$ | Discover useful structure in $X$ |
| **Typical tasks** | Classification, regression | Clustering, dimensionality reduction |

## Common unsupervised tasks

- **Clustering:** Group similar examples.
- **Dimensionality reduction:** Represent data using fewer dimensions.
- **Visualization:** Explore hidden patterns.
- **Anomaly detection:** Identify unusual observations.
- **Representation learning:** Discover useful features.

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> **Key distinction.** Unsupervised learning has **no ground-truth targets for training**, but it still has a learning objective, such as grouping similar examples.

# Why Unsupervised Learning Matters

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# Many datasets are unlabeled

**Labels** can be expensive, slow, subjective, or unavailable.

## Examples

| Domain | Unlabeled data | Possible structure |
|---|---|---|
| Customers | purchase behavior | customer segments |
| Documents | text vectors | topics or themes |
| Images | visual features | similar image groups |
| Students | engagement measures | learning behavior groups |
| Sensors | time-window features | operating modes |
| Network traffic | connection patterns | unusual activity |

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A company may not know customer types in advance, but **clustering** can suggest meaningful segments.

# What Unsupervised Learning Does Not Guarantee

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# Discovered structure needs interpretation

**Unsupervised learning** can discover patterns even when the underlying structure is weak or not meaningful.

## Important cautions

- **Feature scaling:** Different scales can change the discovered clusters.
- **Feature selection:** Irrelevant features may produce misleading patterns.
- **Algorithm choice:** Different methods or settings may find different structures.
- **Number of clusters:** The choice of $K$ can affect the grouping.
- **Interpretation:** A cluster does not automatically represent a meaningful real-world category.

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## A question to ask

> Do these groups represent useful, stable patterns—or are they artifacts of the data and modeling choices?

> **Key lesson.** Treat unsupervised results as **hypotheses to investigate**, not established facts. Check their stability, usefulness, and meaning in the application context.

# Typical Unsupervised Workflow

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# Similar pipeline, different evaluation

Unsupervised learning follows a familiar workflow, but **evaluating the results requires a different approach**.

### 1. Prepare

- Define the analysis question.
- Select relevant features.
- Preprocess and scale data.

### 2. Discover

- Apply clustering or dimensionality reduction.
- Inspect and visualize the results.

### 3. Evaluate

- Interpret clusters or components.
- Check stability and usefulness.
- Use domain knowledge or downstream tasks.

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**Key difference:** Without known target labels, we cannot directly measure predictive accuracy as in supervised learning.

💡 **Key idea.** Unsupervised evaluation focuses on whether the discovered structure is **stable, meaningful, and useful**, not simply whether a prediction matches a known label.

# Part II

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# Clustering

**Clustering** groups observations so that examples in the same group are more similar to each other than to examples in other groups.

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# What is Clustering?

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# Grouping without predefined labels

**Clustering** receives only feature vectors:

$$x_1, x_2, \ldots, x_n$$

It assigns each observation to a cluster:

$$c_i \in \{1, \ldots, K\}$$

## Examples

| Problem | Possible clusters |
|---|---|
| Customer segmentation | budget, regular, premium |
| Student behavior | highly engaged, moderate, inactive |
| Documents | sports, politics, technology |
| Patients | symptom profiles |
| Transactions | normal patterns and unusual patterns |

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> Cluster names are not learned automatically. Humans interpret **clusters** after examining their characteristics.

# Similarity and Distance

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# What does “near” mean?

Many clustering algorithms group observations based on a **distance measure**.

For numerical feature vectors, a common choice is **Euclidean distance**:

$$d(x, z) = \sqrt{\sum_{j=1}^{p} (x_j - z_j)^2}$$

where $x$ and $z$ are observations with $p$ features.

## How to interpret distance

- **Small distance:** Observations are close under the chosen features.
- **Large distance:** Observations are far apart under those features.

However, numerical closeness does not always imply **meaningful real-world similarity**.

> **Feature scaling matters.** As with KNN, features with larger numerical ranges can dominate Euclidean distance unless appropriately scaled.

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# Why Scaling Matters

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# Distance depends on feature units

Consider two customer features used for clustering:

| Feature | Illustrative range |
|---|---|
| Weekly activity count | Approximately 1–9 |
| Annual spending (thousand SAR) | 20–120 |

Without scaling, differences in **spending** may dominate Euclidean distances, even if **activity** contains more useful grouping information.

## Standardization

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$$z = \frac{x - \mu}{\sigma}$$

This transforms each feature to have mean **0** and standard deviation **1** in the data used to fit the scaler.

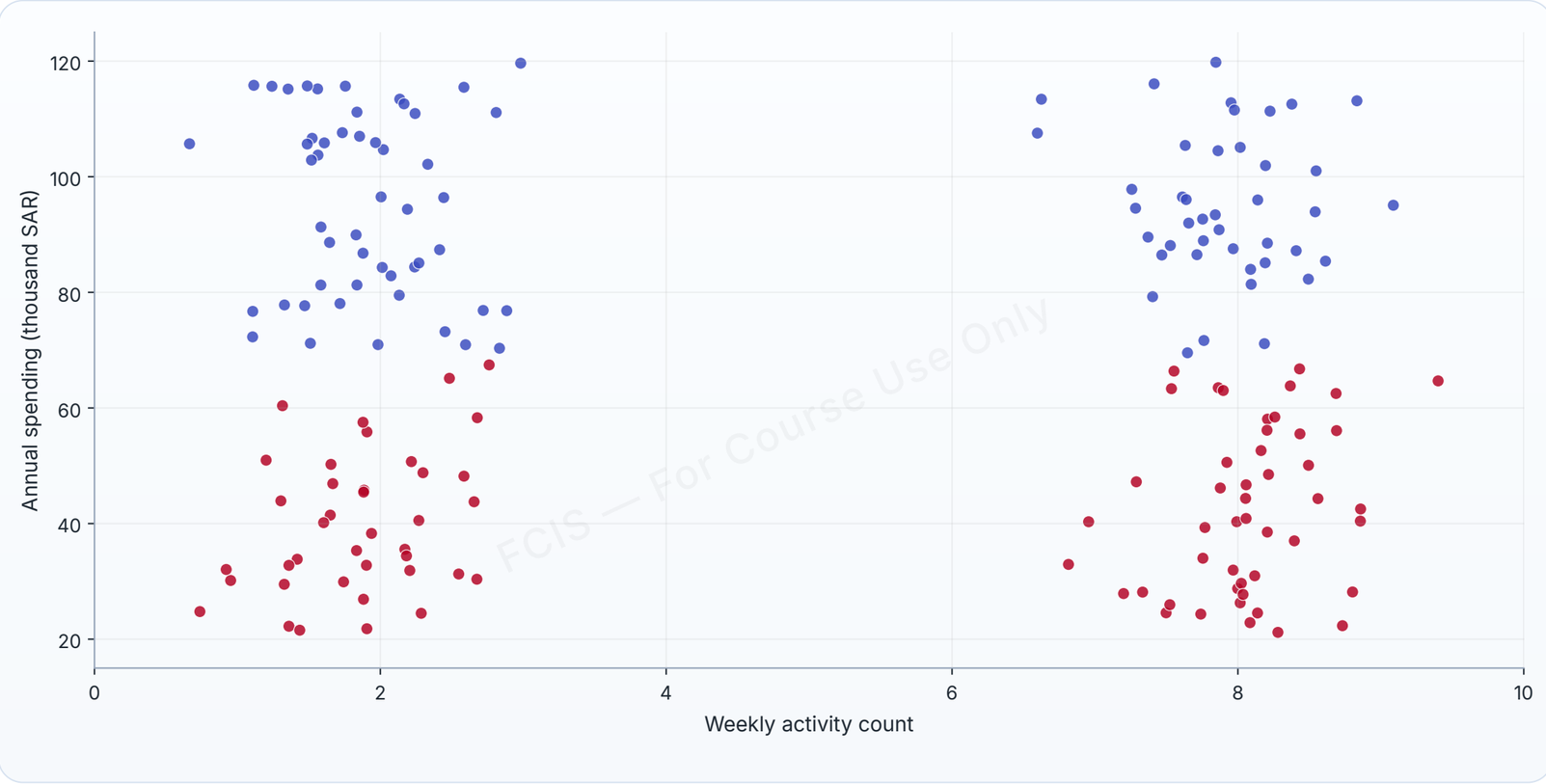
> 💡 **Key idea.** Scaling changes what counts as **near**. Fit the scaler on the data being clustered; if evaluating on held-out data, fit it only on the relevant training partition.

# Plot: K-Means Before Scaling

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# Large numerical ranges can dominate clustering

**Synthetic example:** Weekly activity contains two distinct groups, while spending varies independently over a wider numerical range. Colors indicate the clusters assigned by K-Means.




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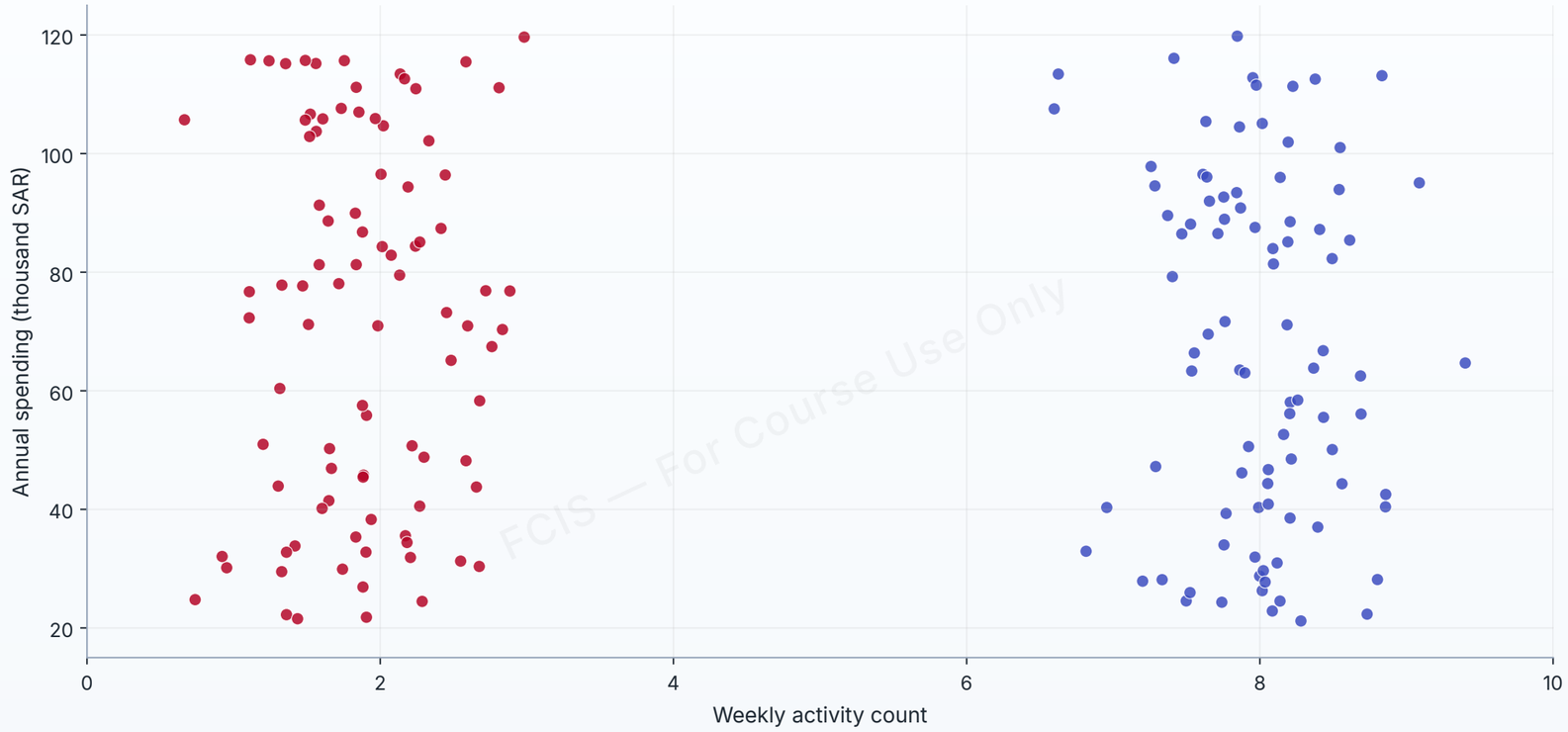
**Observe:** K-Means mainly separates customers by **spending**, producing approximately horizontal color bands. The more distinct activity groups are not recovered.

# Plot: K-Means After Scaling

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# Standardization changes the discovered groups

The **same observations** are clustered again after standardization. The plot still shows the original units so the results can be compared directly.




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💡 **Key idea.** After scaling, K-Means recovers the **low-activity and high-activity groups**. The observations are unchanged—only the distances used for clustering have changed.

# Part III

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# K-Means Clustering

**K-Means** groups similar observations into $K$ clusters, each represented by a **centroid**.

It repeatedly **assigns points** to their nearest centroid and **updates centroids** to the mean of their assigned points.

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# K-Means Intuition

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# Centroids and assignments

**K-Means** partitions $n$ observations into $K$ clusters.

- Each cluster is represented by a **centroid** $\mu_k$.
- Each observation $x_i$ belongs to a cluster $c_i$.
- The objective is to keep observations **close to their assigned centroids**.

## K-Means objective

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$$J = \sum_{i=1}^{n} \|x_i - \mu_{c_i}\|^2$$

This is the **sum of squared distances** from each point to its assigned centroid.

> **Key idea.** K-Means searches for cluster assignments and centroid positions that minimize **within-cluster variation**.

# Animated K-Means Walkthrough

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# Assign points, then update centroids

Consider **six observations** and $K = 2$ clusters. Advance to reveal the **assignment step**, followed by the **centroid update**.

## Six points, two clusters



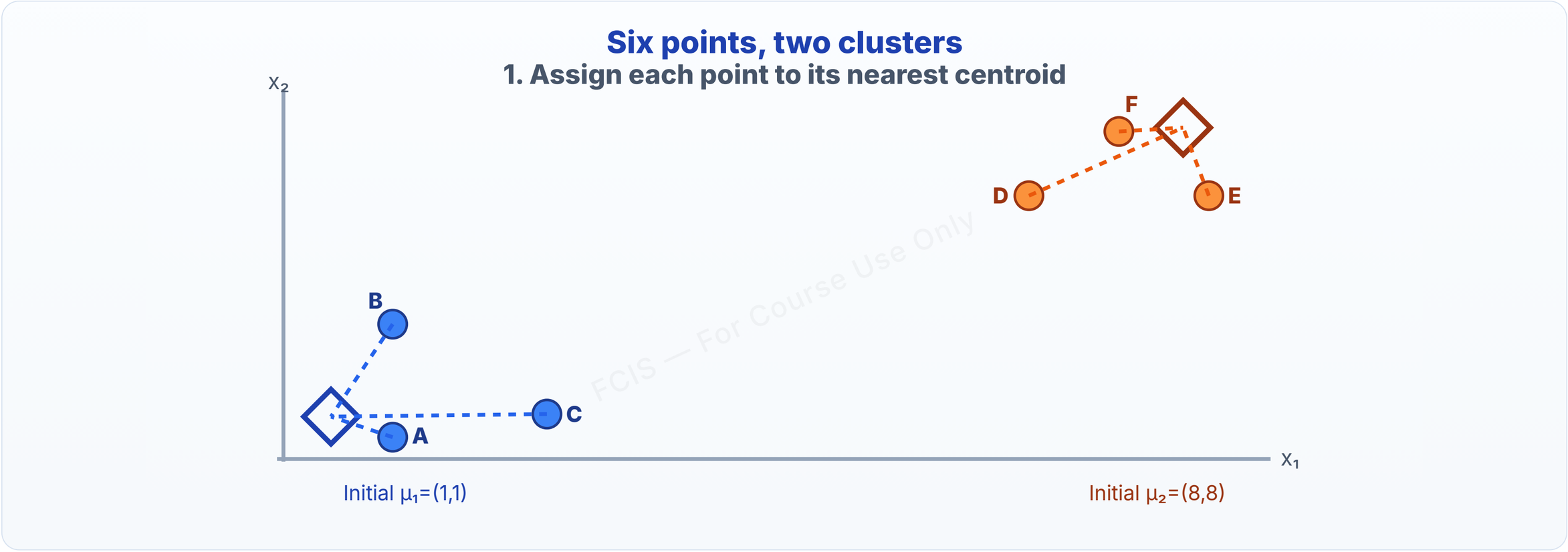

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**Diamonds:** Initial centroids · **Dashed lines:** Assignments · **Green dots:** Updated centroids

💡 **Two repeating steps:** Assign each point to its **nearest centroid**, then move each centroid to the **mean of its assigned points**.

# Assign points, then update centroids

Consider **six observations** and $K = 2$ clusters. Advance to reveal the **assignment step**, followed by the **centroid update**.



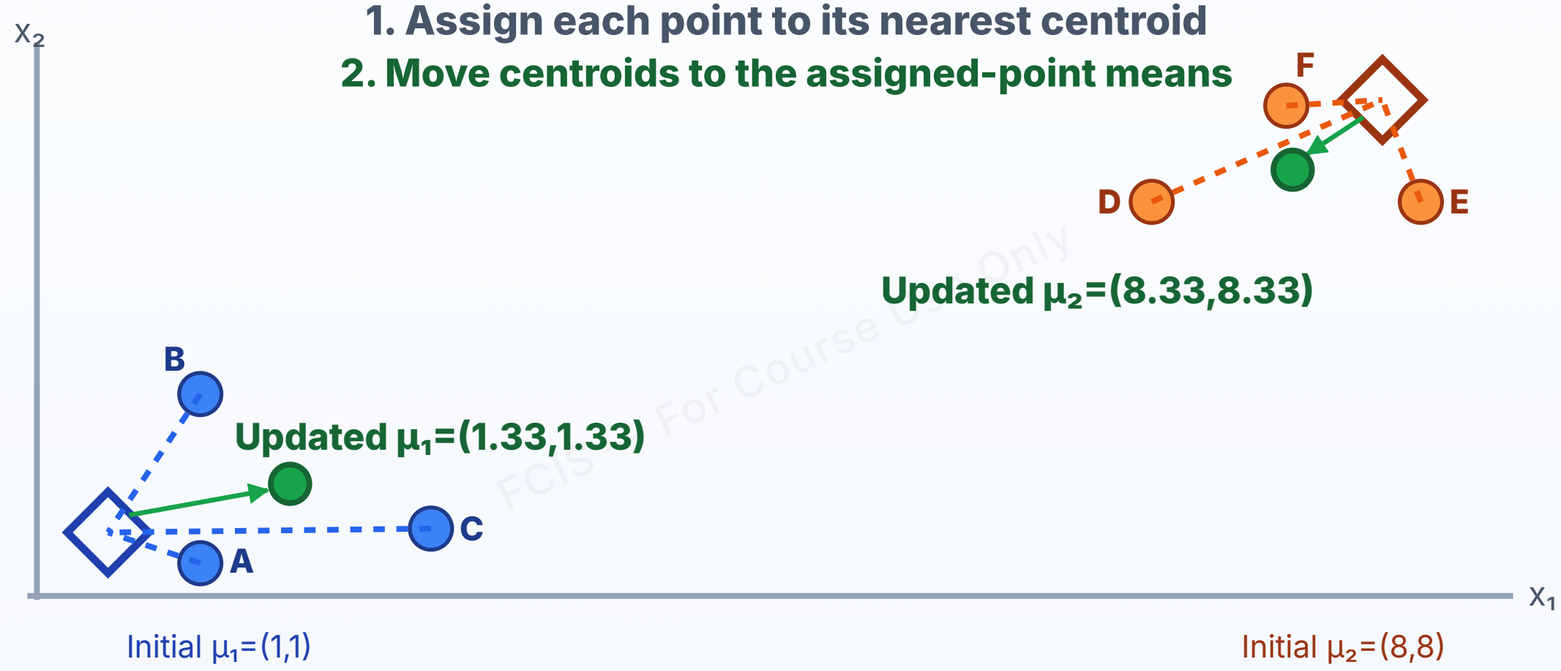

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# K-Means Algorithm

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# Assignment and update

K-Means alternates between two operations until the clusters stabilize.

```
Choose the number of clusters K
Initialize K centroids

repeat:
    1. Assignment:
       Assign each point to its nearest centroid

    2. Update:
       Replace each centroid with the mean
       of its assigned points

until assignments stabilize or a stopping limit is reached
```

## Important details

- The **assignments** depend on current centroid positions.
- The **centroids** depend on current cluster assignments.
- Different initial centroids can lead to different solutions.

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💡 **Key idea.** K-Means repeatedly improves its clustering, but the result can depend on **initialization**. It is not guaranteed to find the global minimum.

# Assignment Step

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# Choose the nearest centroid

Given the current centroids $\mu_1, \ldots, \mu_K$, assign each observation $x_i$ to the cluster with the **smallest squared Euclidean distance**:

$$c_i = \arg\min_{k \in \{1,\ldots,K\}} \|x_i - \mu_k\|^2$$

## What does this mean?

- Calculate the distance from $x_i$ to each centroid.
- Find the **closest centroid**.
- Assign $x_i$ to that centroid's cluster.

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**Example.** If a point is closer to $\mu_1$ than to $\mu_2$, it is assigned to **cluster 1**. The centroid positions remain unchanged during this step.

# Update Step

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# Move centroids to cluster means

After assigning observations, recompute each **centroid** as the mean of the points in its cluster:

$$\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

Here, $C_k$ is the set of points assigned to cluster $k$, and $|C_k|$ is the number of points.

## Example: The first cluster

Suppose the assigned points are $A = (1, 1)$, $B = (1, 2)$, and $C = (2, 1)$.

$$\mu_1 = \left( \frac{1+1+2}{3}, \frac{1+2+1}{3} \right) \approx \boxed{(1.33,\ 1.33)}$$

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💡 **Key idea.** The centroid moves to the **average position** of its assigned observations. K-Means then repeats the assignment step using the updated centroids.

# Worked K-Means Example

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# Dataset and initial centroids

Consider **six observations** in two dimensions:

| Point | Coordinates |
|---|---|
| A | $(1, 1)$ |
| B | $(1, 2)$ |
| C | $(2, 1)$ |
| D | $(8, 8)$ |
| E | $(9, 8)$ |
| F | $(8, 9)$ |

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We choose $K = 2$ clusters, with initial centroids:

$$\mu_1 = (1, 1), \qquad \mu_2 = (8, 8)$$

**Task.** Assign each observation to its **nearest centroid**, then update the centroid positions.

# Assignment Step Example

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# Compute distances to the centroids

For each observation, compare its distance to $\mu_1 = (1, 1)$ and $\mu_2 = (8, 8)$.

| Point | $d(x, \mu_1)$ | $d(x, \mu_2)$ | Assigned cluster |
|---|---|---|---|
| A $(1, 1)$ | **0** | $\sqrt{98}$ | $C_1$ |
| B $(1, 2)$ | **1** | $\sqrt{85}$ | $C_1$ |
| C $(2, 1)$ | **1** | $\sqrt{85}$ | $C_1$ |
| D $(8, 8)$ | $\sqrt{98}$ | **0** | $C_2$ |
| E $(9, 8)$ | $\sqrt{113}$ | **1** | $C_2$ |
| F $(8, 9)$ | $\sqrt{113}$ | **1** | $C_2$ |

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## Resulting clusters

$$\boxed{C_1 = \{A, B, C\}} \qquad \boxed{C_2 = \{D, E, F\}}$$

💡 **Assignment step:** Each point joins its **nearest centroid**. We can compare squared distances instead of taking square roots; the assignments are identical.

# Update Step Example

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# Recompute the centroid positions

After assigning the points, each centroid moves to the **mean of its cluster**.

**Cluster** $C_1 = \{A, B, C\}$

$$\mu_1 = \frac{(1,1) + (1,2) + (2,1)}{3}$$
$$= \left(\frac{4}{3}, \frac{4}{3}\right) \approx \boxed{(1.33,\ 1.33)}$$

**Cluster** $C_2 = \{D, E, F\}$

$$\mu_2 = \frac{(8,8) + (9,8) + (8,9)}{3}$$
$$= \left(\frac{25}{3}, \frac{25}{3}\right) \approx \boxed{(8.33,\ 8.33)}$$

**Update step:** Both centroids move toward the **average position** of their assigned observations.

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# Objective Check After One K-Means Update

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# Did the clustering objective improve?

**WCSS** measures the total squared distance between observations and their assigned centroids.

## Compare before and after the update

Using the **same cluster assignments**:

$$WCSS_{\text{before}} = (0 + 1 + 1) + (0 + 1 + 1) = \boxed{4}$$

$$WCSS_{\text{after}} = \frac{4}{3} + \frac{4}{3} = \boxed{\frac{8}{3} \approx 2.67}$$

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The assignments are unchanged, but the updated centroids are **closer to their assigned points overall**.

> 💡 **Why does WCSS decrease?** For fixed cluster assignments, the **mean minimizes the sum of squared distances**. Therefore, updating centroids cannot increase WCSS.

# Continue Until Convergence

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# When does K-Means stop?

After updating the centroids, K-Means repeats:

1. **Assign:** Find the nearest updated centroid for each point.
2. **Update:** Recalculate the mean of each cluster.
3. **Check:** Determine whether the clustering has stabilized.

## In our worked example

- The updated centroids are approximately $(1.33, 1.33)$ and $(8.33, 8.33)$.
- Reassigning the six points produces the **same two clusters**.
- Another centroid update would produce **no change**.

The algorithm has therefore **converged**.

> **Important.** K-Means may stop when assignments stabilize, centroid movement is sufficiently small, or an iteration limit is reached. Convergence does not guarantee the **globally best clustering**.

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# K-Means Objective

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# Within-cluster sum of squares (WCSS)

K-Means minimizes the **within-cluster sum of squares**:

$$WCSS = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Here:

- $C_k$: Set of observations assigned to cluster $k$.
- $\mu_k$: Mean (centroid) of cluster $k$.
- $\|x_i - \mu_k\|^2$: Squared distance to that centroid.

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## How should we interpret WCSS?

- **Lower WCSS:** More compact clusters under the chosen distance.
- **Larger $K$:** Typically produces lower WCSS because more centroids are available.

**Key caution.** A lower WCSS does not automatically mean **better or more meaningful clusters**. WCSS alone is not sufficient for choosing $K$.

# Why Initialization Matters

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# Different starts can give different results

K-Means begins with **initial centroid positions**, which influence the entire clustering process.

## Poor initialization can cause

- Convergence to a **poor local optimum**.
- **Slow convergence** over many iterations.
- Different or unstable cluster assignments.
- **Empty clusters**, which require special handling.

## How can we improve initialization?

- **Multiple runs:** Try different initial centroids and keep the solution with the lowest WCSS.
- **K-Means++:** Choose initial centroids more carefully, favoring points far from those already selected.

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💡 **Key idea.** K-Means is sensitive to its starting centroids. Multiple initializations and **K-Means++** can improve results, but neither guarantees the global optimum.

# Choosing the Number of Clusters

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# How do we select $K$?

K-Means requires the number of clusters $K$ to be specified before fitting.

## Common approaches

- **Domain knowledge:** How many groups would be meaningful?
- **Elbow method:** Where does the WCSS improvement begin to level off?
- **Silhouette score:** How well separated are the clusters?
- **Cluster stability:** Are similar groups found across repeated analyses?
- **Downstream usefulness:** Do the clusters improve later decisions?
- **Practical requirements:** How many groups can the application use?

**Example.** A university may prefer **three interpretable student-support profiles** if they lead to useful interventions, even when a different $K$ gives a slightly better numerical score.

💡 **Key idea.** Choose $K$ using both **quantitative evidence** and the **meaning and usefulness** of the resulting clusters.

# Elbow Method

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# Look for diminishing returns

The **elbow method** plots the **within-cluster sum of squares (WCSS)** against the number of clusters $K$.

- As $K$ increases, WCSS generally **decreases**.
- More centroids can fit the data more closely.
- We look for a point where adding clusters produces only **small further reductions**.

## How to interpret the elbow

The bend in the curve suggests a possible balance between:

- **Compactness:** Keeping points close to their centroids.
- **Simplicity:** Avoiding unnecessarily many clusters.

**Important.** The elbow is a **heuristic**, not a guaranteed optimal choice. Some datasets have no clear elbow, so also consider cluster stability and usefulness.

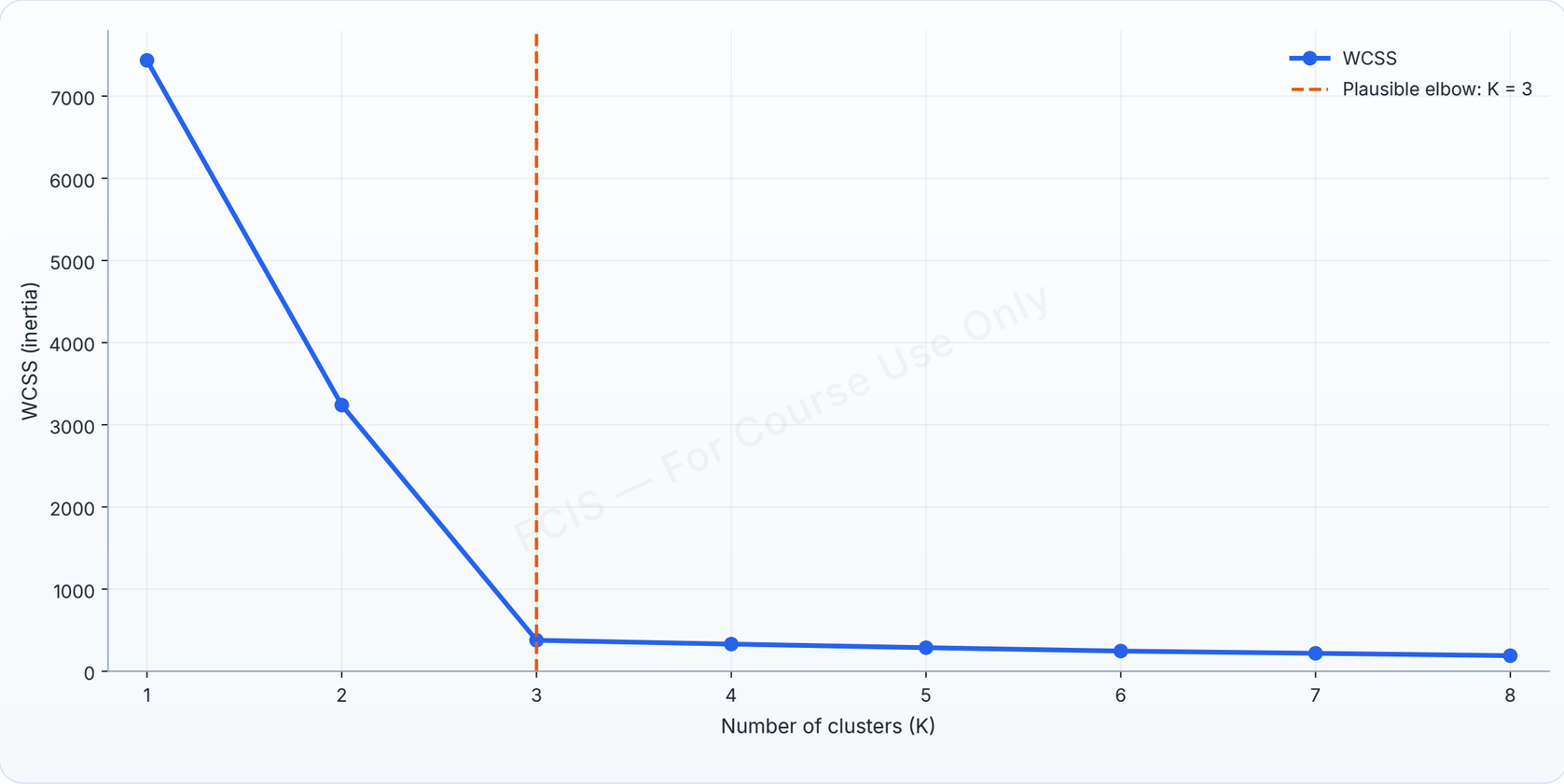
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# Plot: K-Means Elbow Curve

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# Where do improvements begin to level off?

The dataset contains **three simulated groups**. We fit K-Means with different values of $K$ and compare their WCSS.




💡 **Interpretation.** WCSS drops sharply at first, then improves more slowly after approximately $K = 3$. The simulated data were generated from three groups, but real datasets may not show such a clear elbow.

# Silhouette Score

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# Measure cohesion and separation

The **silhouette score** evaluates whether observations are closer to their own cluster than to other clusters.

For each observation $i$:

- $a_i$**:** Average distance to other points in its own cluster.
- $b_i$**:** Lowest average distance to points in another cluster.

## Silhouette value

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

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## Interpretation

| Silhouette value | Meaning |
|---|---|
| **Near +1** | Well matched to its own cluster |
| **Near 0** | Near a cluster boundary |
| **Negative** | May fit another cluster better |

💡 **Key idea.** Silhouette values range from **−1 to +1**. The average silhouette score helps compare clustering solutions, but it does not guarantee meaningful groups.

# K-Means Assumptions

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# What kinds of clusters does it prefer?

**K-Means** represents clusters using their **mean centroids** and minimizes squared Euclidean distances.

## Works well when

- Clusters are **compact and roughly spherical**.
- Clusters have similar **spread and density**.
- Groups are reasonably **separated**.
- Means provide useful cluster representatives.

## May struggle with

- **Curved or elongated** clusters.
- Groups with very different densities or sizes.
- **Outliers** that distort centroids.
- Categorical features without suitable encoding.
- High-dimensional data with irrelevant features.

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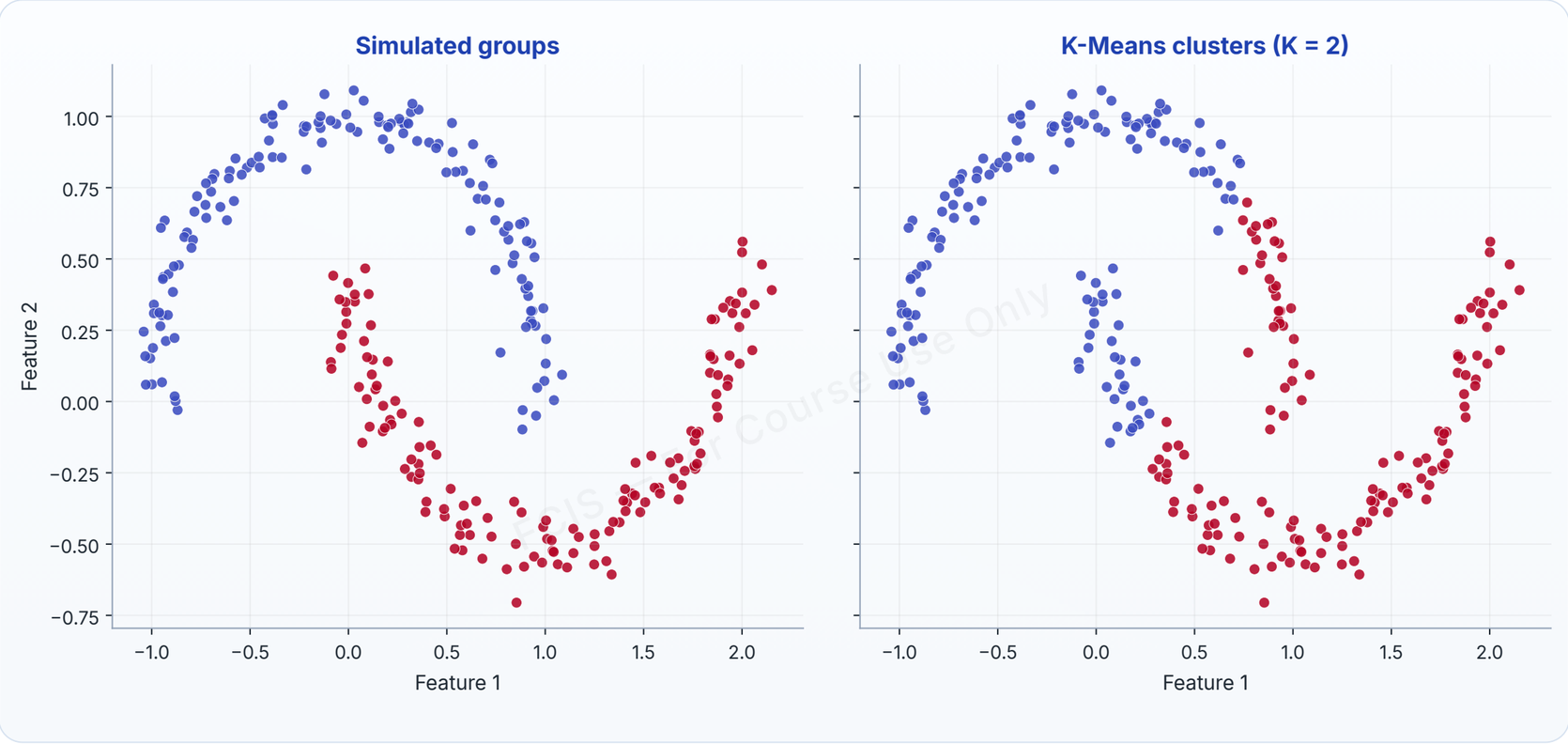
**Key limitation.** K-Means assigns points to the nearest centroid, creating **convex cluster regions**. It may not recover complex shapes, even when the groups are visually obvious.

# Plot: Where K-Means Can Struggle

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# Curved groups are difficult for centroid-based clustering

The same **two-moons dataset** is shown twice. Left: the original simulated groups. Right: the clusters found by K-Means with $K = 2$.




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💡 **Key idea.** The original groups form two **curved moons**, but K-Means partitions space according to the **nearest centroid**. Choosing the correct $K$ does not guarantee recovering the true group structure.

# Interpreting Clusters

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# Clusters need meaning, not just numbers

After fitting a clustering model, examine **what distinguishes each group**.

## Questions to investigate

1. **Centroids:** What are the typical feature values?
2. **Distinctive features:** How does this cluster differ from others?
3. **Size:** Is the group large enough to be meaningful?
4. **Stability:** Does a similar group appear across repeated analyses?
5. **Domain knowledge:** Does the grouping make sense in context?
6. **Usefulness:** Can the cluster support a meaningful decision or action?

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## Example

A customer cluster with **frequent purchases** and **high activity** might be described as a *high-engagement segment*.

> **Important.** Cluster labels are **interpretations made after examining the data**. K-Means discovers numerical groupings; it does not assign real-world meaning automatically.

# K-Means Mini-Case

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# Customer segmentation

A store uses **K-Means ($K = 4$)** to group customers based on purchasing behavior.

**Features:** Annual spending · Number of visits · Recency of last purchase · Average basket size

## Possible cluster profiles

| Cluster | Observed pattern | Possible interpretation |
|---|---|---|
| **C1** | Frequent visits, low spending | Frequent low-spending customers |
| **C2** | Frequent visits, high spending | High-value loyal customers |
| **C3** | Long time since last purchase | Inactive customers |
| **C4** | Few visits, large baskets | Occasional large-purchase customers |

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## From clusters to decisions

The store can investigate which groups may benefit from **loyalty offers, re-engagement campaigns, or tailored promotions**.

**Important.** K-Means finds numerical groups, **not customer categories**. Interpret centroid patterns, check cluster stability, and assess usefulness before acting.

# Part IV

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# Hierarchical Clustering

**Hierarchical clustering** organizes observations into a **tree of nested groups**.

Unlike K-Means, it does not require choosing the final number of clusters before building the hierarchy.

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# Hierarchical Clustering Intuition

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# Building a tree of clusters

**Agglomerative hierarchical clustering** starts with each observation in its own cluster and repeatedly merges groups.

## How it works

1. **Start:** Treat each of the $n$ observations as a separate cluster.
2. **Compare:** Measure distances between clusters using a chosen linkage rule.
3. **Merge:** Combine the two closest clusters.
4. **Repeat:** Continue until all observations belong to one cluster.

The sequence of merges produces a tree called a **dendrogram**.

> 💡 **Key idea.** The dendrogram preserves the **nested grouping structure**. Cutting it at different heights produces different numbers of clusters.

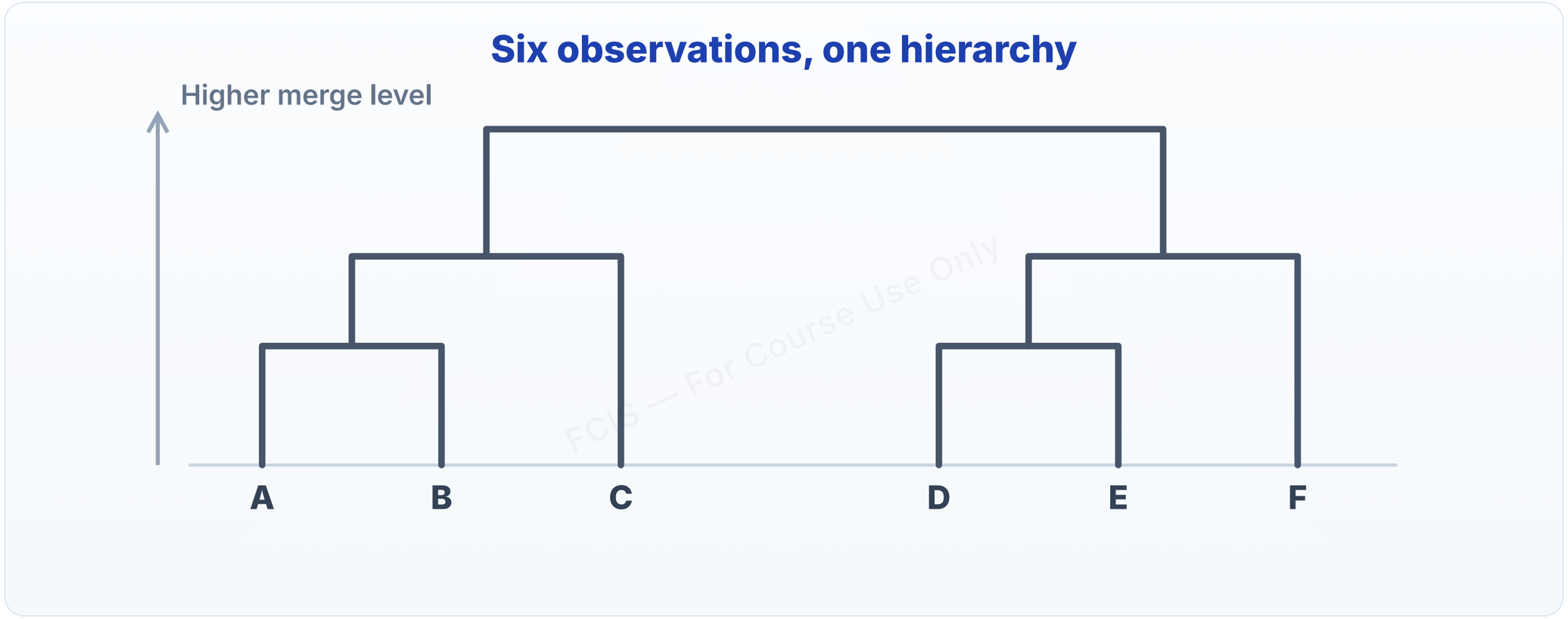
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# Dendrogram Walkthrough

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# Cutting the hierarchy into clusters

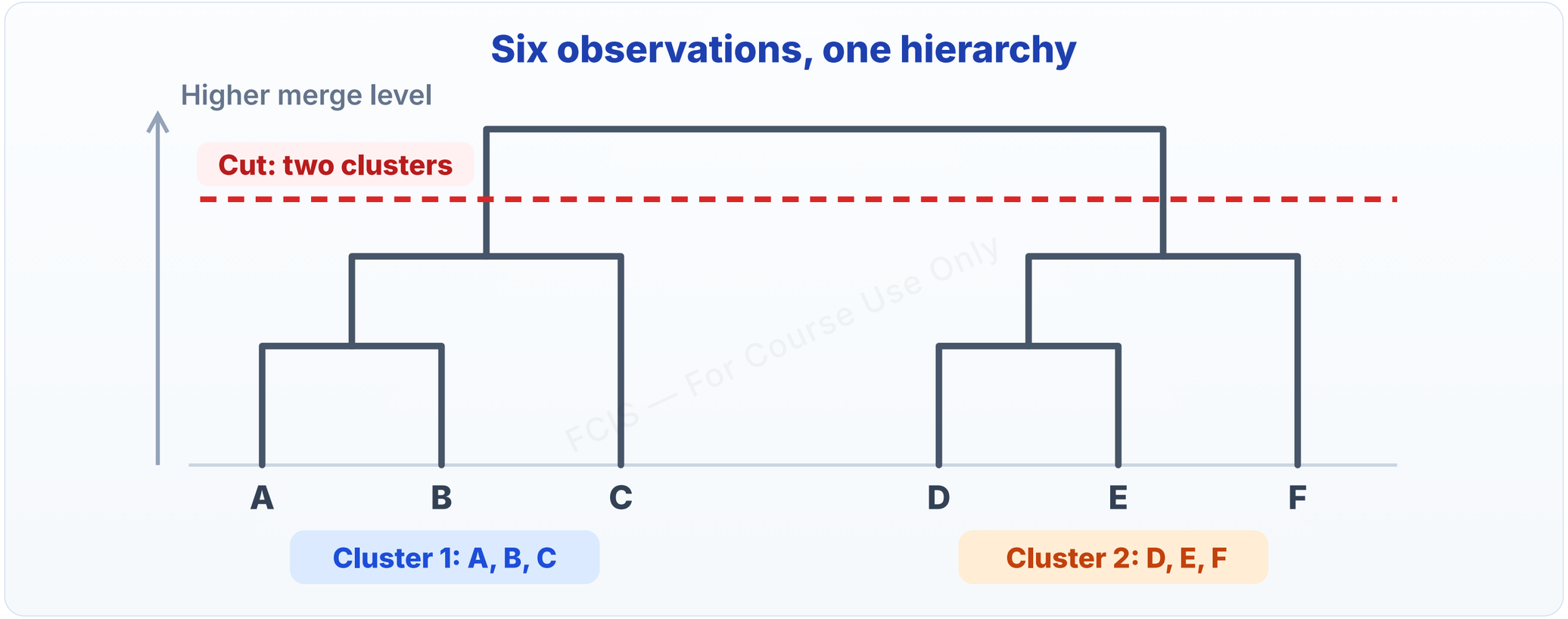
The **dendrogram** shows which groups merge and at what height. Higher merges indicate greater dissimilarity under the chosen linkage rule.




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# Cutting the hierarchy into clusters

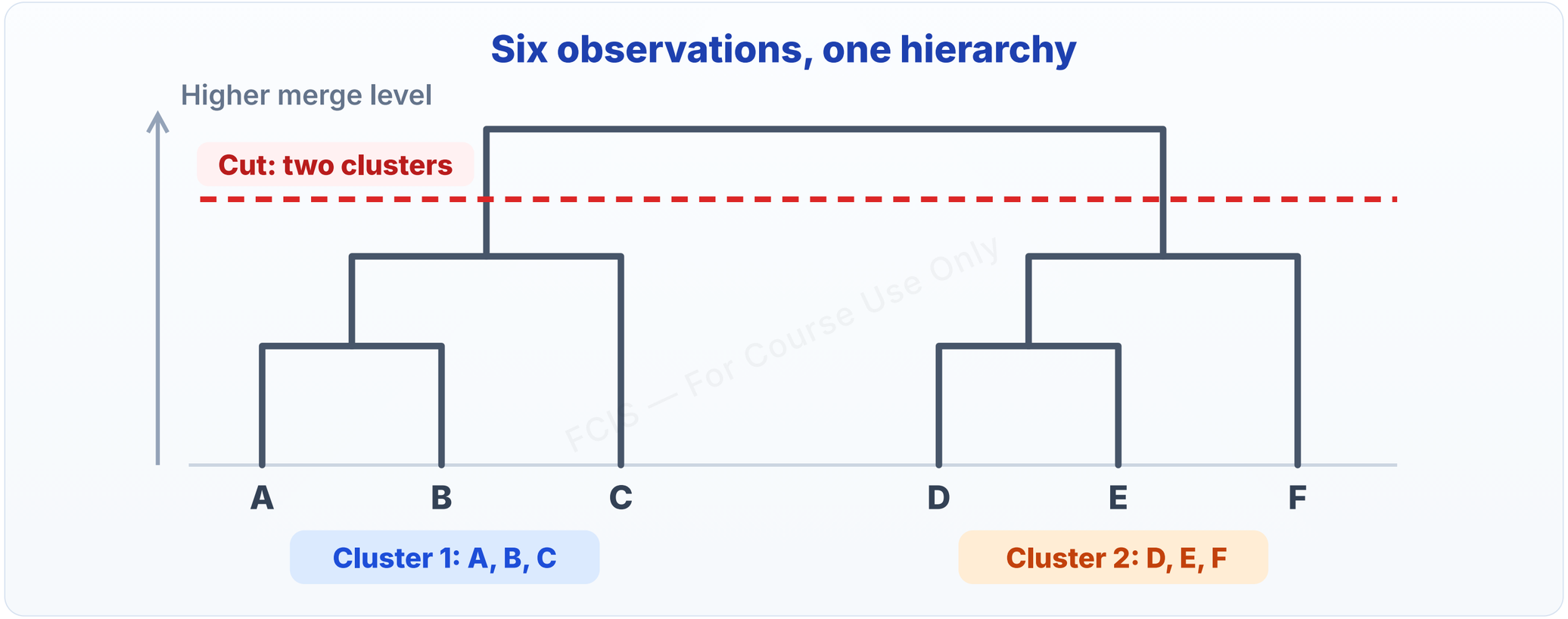
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# Cutting the hierarchy into clusters

The **dendrogram** shows which groups merge and at what height. Higher merges indicate greater dissimilarity under the chosen linkage rule.




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💡 **Observe:** Cutting **below the final merge** gives two clusters: $\{A, B, C\}$ and $\{D, E, F\}$. A lower cut could produce more groups. Merge heights are illustrative.

# Linkage Criteria

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# How do we measure distance between clusters?

Hierarchical clustering needs a **linkage rule** to decide which clusters should merge next.

| Linkage | How it chooses clusters to merge |
|---|---|
| **Single** | Smallest distance between any two points |
| **Complete** | Smallest maximum pairwise distance |
| **Average** | Smallest average pairwise distance |
| **Ward** | Smallest increase in within-cluster squared error |

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## Why does linkage matter?

Different rules may discover different structures:

- **Single linkage:** Can connect groups through chains of nearby points.
- **Complete linkage:** Tends to form compact groups.
- **Average linkage:** Balances distances across cluster members.
- **Ward linkage:** Favors groups with low within-cluster variation.

💡 **Key idea.** The **linkage criterion** determines how clusters are compared, so different linkage choices can produce different dendrograms.

# Dendrogram Interpretation

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# How do we choose the number of clusters?

A **dendrogram** displays the order in which clusters merge and their associated merge heights.

## Reading the dendrogram

1. **Leaves:** Individual observations at the bottom.
2. **Branches:** Connections showing which groups merge.
3. **Merge height:** The distance or dissimilarity level of a merge.
4. **Horizontal cut:** The number of branches crossed gives the number of clusters.

## Choosing a cut height

- A **lower cut** generally gives **more clusters**.
- A **higher cut** generally gives **fewer clusters**.
- Large gaps between merge heights may suggest useful separation.

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**Example.** Cutting our six-point dendrogram below its final merge yields **two clusters**. Cutting lower could separate the smaller subgroups.

A large gap is a useful visual clue, **not proof** that the corresponding clusters are meaningful.

# K-Means vs. Hierarchical Clustering

For Course Only

# Two different ways to discover groups

| Aspect | K-Means | Hierarchical clustering |
|---|---|---|
| **Choose $K$ initially?** | Yes | Not required to build dendrogram |
| **Output** | One partition | Nested hierarchy and possible partitions |
| **Grouping rule** | Nearest centroid | Chosen distance and linkage |
| **Cluster shape** | Favors compact, convex groups | Depends on linkage |
| **Scalability** | Usually better for large datasets | Often more computationally demanding |
| **Initialization** | Sensitive to starting centroids | No random centroids; ties may matter |
| **Interpretation** | Centroid profiles | Merge history and nested groups |

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💡 **Key idea.** Use **K-Means** when a compact partition is the goal; consider **hierarchical clustering** when understanding nested relationships and exploring different numbers of groups is valuable.

# Part V

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# Dimensionality Reduction

**Dimensionality reduction** represents data using **fewer features** while trying to preserve important patterns and variation.

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# Why Reduce Dimensionality?

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# Too many features can be difficult

High-dimensional data can create several challenges:

- **Visualization:** Difficult to inspect many features together.
- **Distance:** Irrelevant dimensions can make similarity less informative.
- **Overfitting:** Too many features may increase model complexity.
- **Computation:** More dimensions can increase processing costs.
- **Redundancy:** Correlated features may repeat similar information.
- **Noise:** Uninformative features may hide useful patterns.

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## The goal

Transform $p$ original features into $q$ features, where $q < p$:

$$\boxed{X \in \mathbb{R}^{n \times p} \longrightarrow Z \in \mathbb{R}^{n \times q}}$$

💡 **Key idea.** Dimensionality reduction makes data more compact, but reducing dimensions can also **discard information**.

# Feature Selection vs. Feature Extraction

For Course Use Only

# Two ways to reduce features

Dimensionality can be reduced by either **selecting existing features** or **creating new ones**.

| Approach | What it does | Example |
|---|---|---|
| **Feature selection** | Keeps a subset of original features | Select 10 of 100 variables |
| **Feature extraction** | Constructs new features from existing ones | PCA components |

## Important distinction

- **Selection:** Original feature meanings are preserved.
- **Extraction:** New features may summarize patterns more efficiently, but can be harder to interpret.

**PCA is feature extraction.** It creates new features called **principal components**, rather than simply choosing a subset of the original variables.

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# What PCA Tries to Do

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# Find directions of maximum variance

**Principal Component Analysis (PCA)** finds new, mutually perpendicular axes called **principal components**.

- **First component ($PC_1$):** Captures the greatest possible variance.
- **Second component ($PC_2$):** Captures the greatest remaining variance while being perpendicular to $PC_1$.
- **Later components:** Capture progressively less variance.

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## Intuition

PCA finds a new coordinate system aligned with the main directions in which the data vary.

> 💡 **Key idea.** PCA is **unsupervised**: it discovers directions of variation without using target labels. Higher variance does not necessarily mean greater predictive usefulness.

# PCA Geometry

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# Project data onto new axes

Suppose we have two **correlated numerical features**, $x_1$ and $x_2$.

PCA finds two new, perpendicular axes:

- $PC_1$: Follows the direction with the greatest spread.
- $PC_2$: Captures variation perpendicular to $PC_1$.

## How reduction works

1. **Center** the observations.
2. Find the principal directions.
3. **Project** observations onto those directions.
4. Keep only the leading components if they preserve enough variation.

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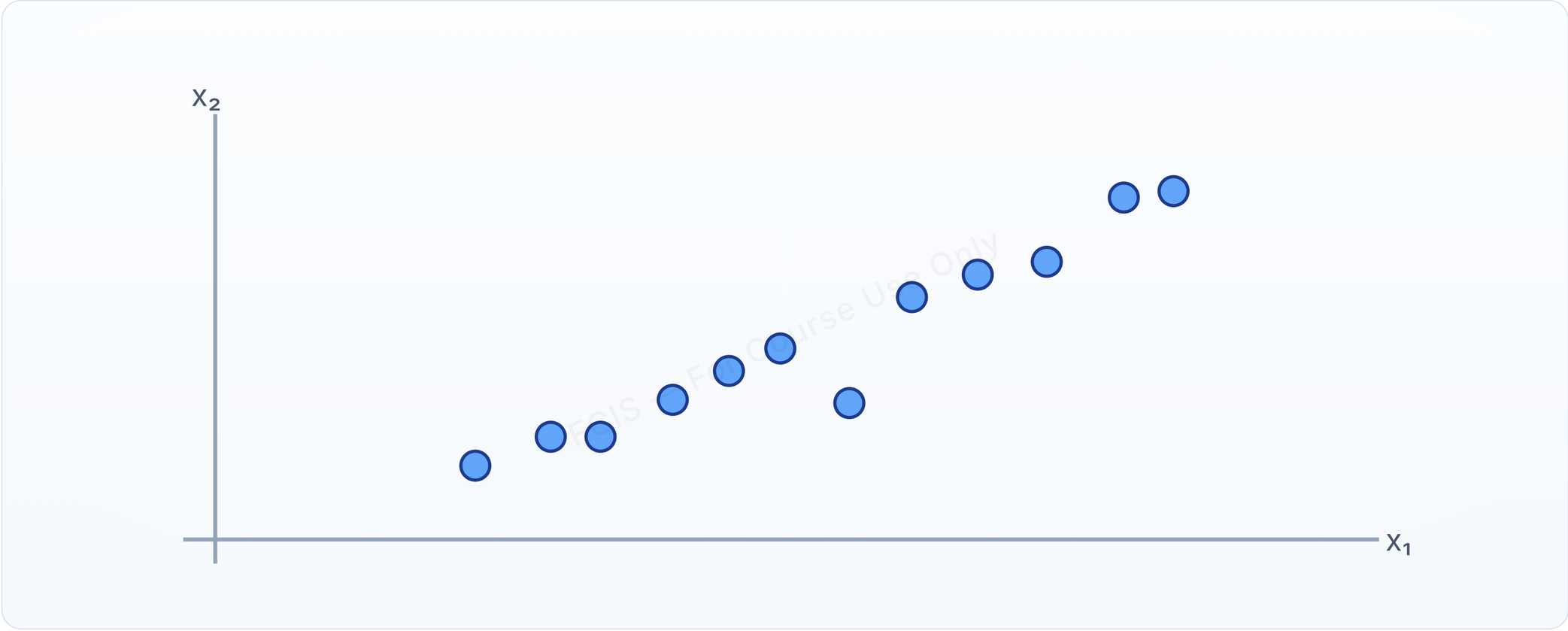
**Example.** If most variation lies along $PC_1$, we can represent the original **2D observations using one component** by keeping $PC_1$ and discarding $PC_2$.

# Animated PCA Projection

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# From original features to principal components

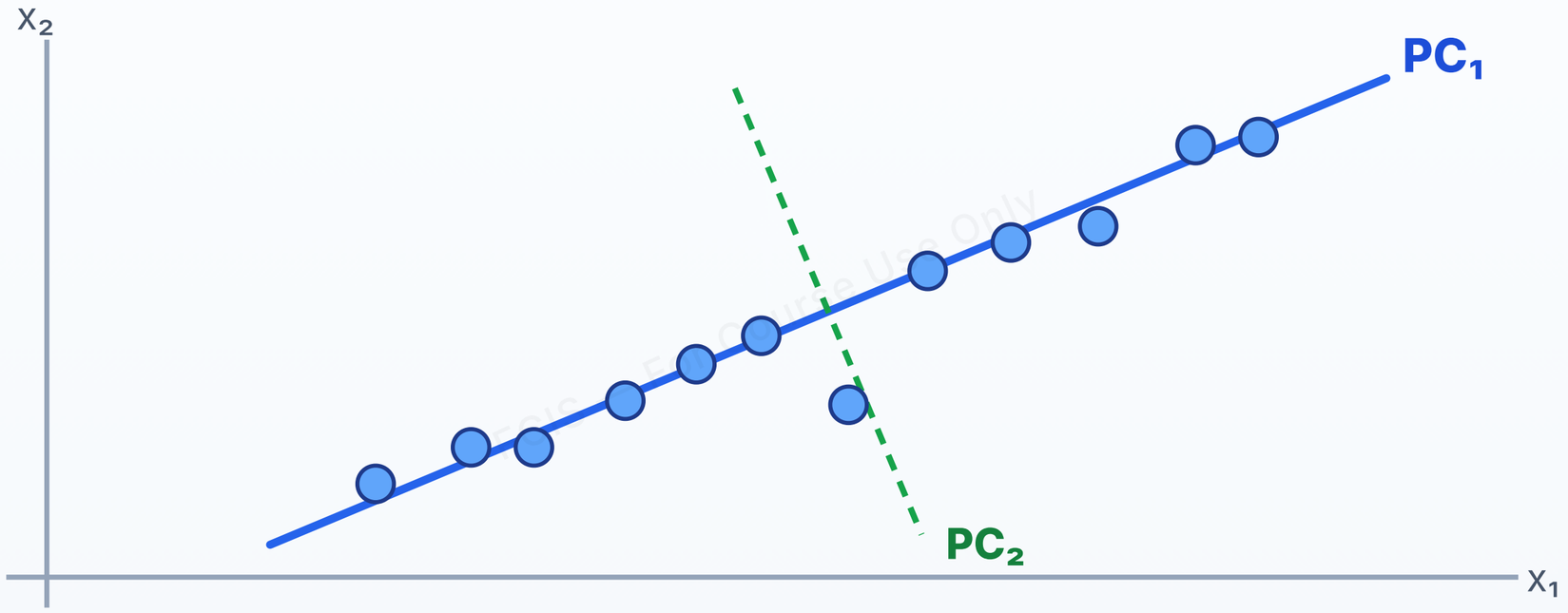
The observations follow a diagonal pattern. Advance to reveal the **principal axes**, then the **projection of one observation onto** $PC_1$.




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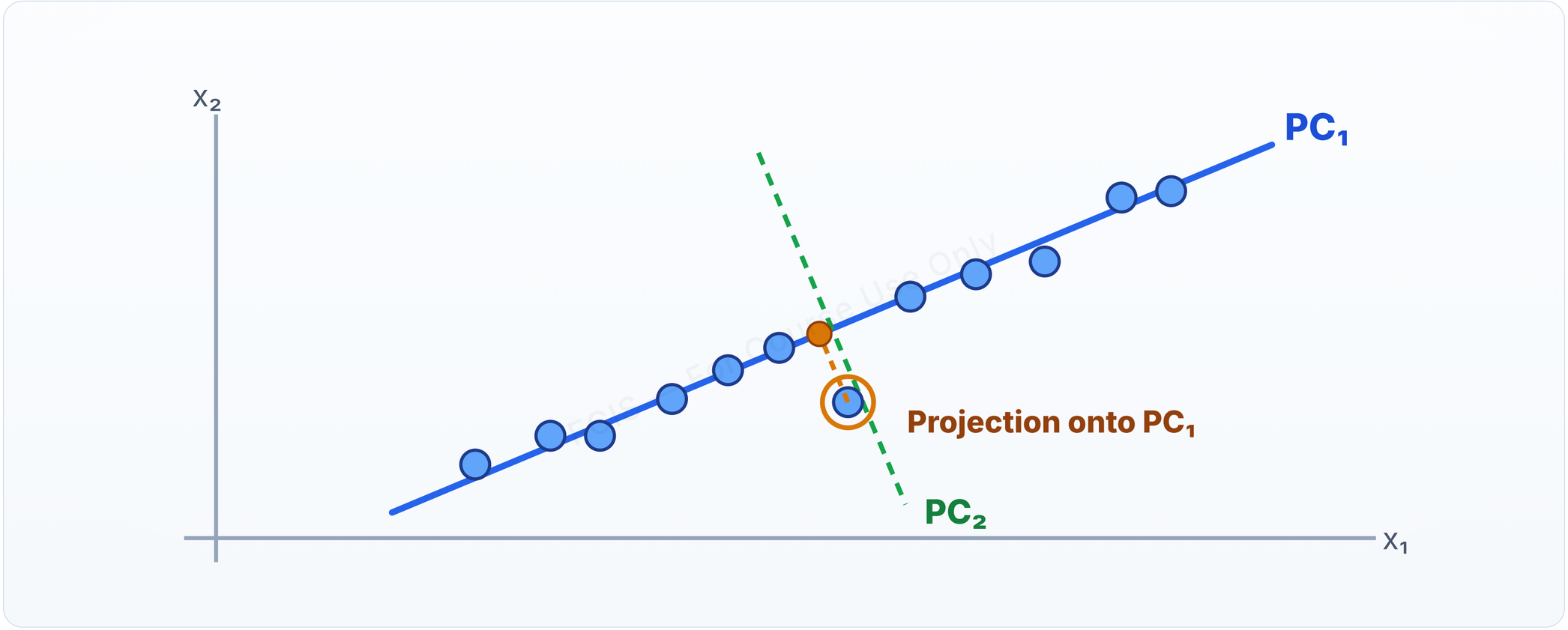
# From original features to principal components

The observations follow a diagonal pattern. Advance to reveal the **principal axes**, then the **projection of one observation onto** $PC_1$.



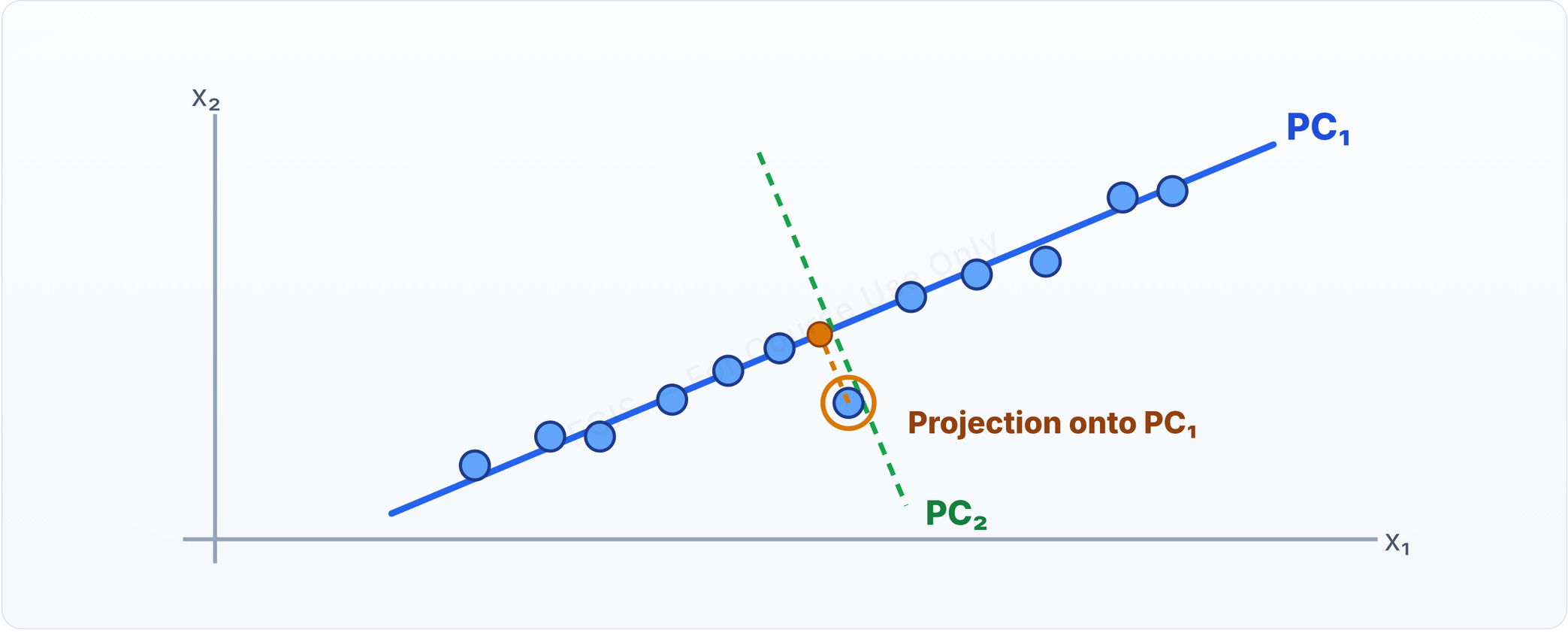
# From original features to principal components

The observations follow a diagonal pattern. Advance to reveal the **principal axes**, then the **projection of one observation onto** $PC_1$.



# From original features to principal components

The observations follow a diagonal pattern. Advance to reveal the **principal axes**, then the **projection of one observation onto** $PC_1$.




💡 **Observe:** Projecting onto $PC_1$ keeps each observation's position along the main direction of variation while discarding variation perpendicular to it.

# Centering and Scaling for PCA

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# Preprocessing matters

PCA depends on **variance**, so feature preprocessing influences the principal components.

## Two important operations

1. **Centering:** Subtract the mean from each feature.
2. **Standardization:** Also divide by the standard deviation when features have different units or scales.

$$z = \frac{x - \mu}{\sigma}$$

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## Why does scaling matter?

Without scaling, features with larger numerical variation may dominate the principal components.

> **Avoid leakage.** When PCA is part of a supervised pipeline, fit **scaling and PCA only on training data** (or the training folds), then transform validation/test data using those fitted transformations.

# Principal Components

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# Linear combinations of original features

Each **principal component** is a direction formed by combining the original features.

For one observation with **centered feature values** $x_1^c, \ldots, x_p^c$, its score on the first component is:

$$\boxed{PC_1 = a_1 x_1^c + a_2 x_2^c + \cdots + a_p x_p^c}$$

The coefficients $a_1, \ldots, a_p$ describe how strongly each original feature contributes to the component direction.

## Interpreting the coefficients

- **Large magnitude:** Strong contribution to the direction.
- **Positive or negative sign:** Direction of the contribution.
- **Small magnitude:** Relatively weak contribution.

> If $PC_1$ gives large weights to **income** and **spending**, it might reflect a general *customer value* direction—an interpretation to check against the data.

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# Explained Variance

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# How much variation does each component capture?

PCA orders components from **highest to lowest explained variance**.

For component $j$:

$$EVR_j = \frac{\lambda_j}{\sum_{k=1}^{p} \lambda_k}$$

where $\lambda_j$ is the variance captured along $PC_j$.

## Cumulative explained variance

The proportion retained by the first $q$ components is:

$$\text{Cumulative EVR}(q) = \sum_{j=1}^{q} EVR_j$$

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💡 **Key idea.** Explained variance helps us decide **how many components to keep**. It measures retained variation, not necessarily retained predictive information.

# Worked PCA Interpretation

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# How many components should we keep?

Suppose PCA produces the following **illustrative explained-variance ratios**:

| Component | Explained variance | Cumulative variance |
|---|---|---|
| $PC_1$ | 52% | 52% |
| $PC_2$ | 25% | 77% |
| $PC_3$ | 12% | **89%** |
| $PC_4$ | 7% | **96%** |
| $PC_5$ | 4% | 100% |

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## Interpretation

- **First 2 components:** Retain **77%** of total variance.
- **First 3 components:** Retain **89%**.
- **First 4 components:** Retain **96%**.

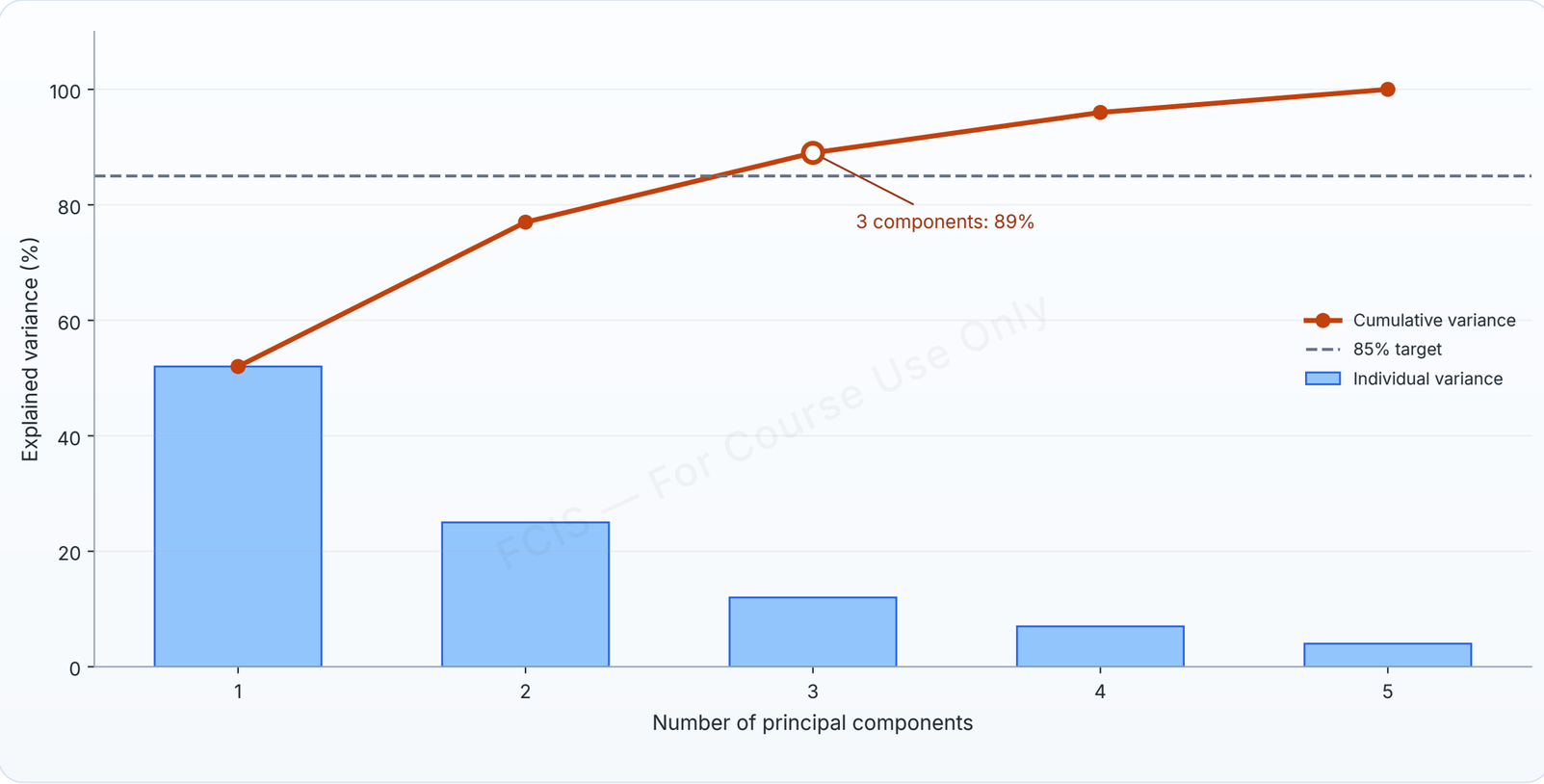
**Decision:** To retain at least **85%** of total variance, keep **3 components**. To retain at least **95%**, keep **4 components**.

# Plot: How Many Principal Components?

For Course Only

# Use cumulative explained variance

The plot uses the **same illustrative values** as the preceding table. The dashed line marks an example target of **85% explained variance**.




💡 **Interpretation.** The first **3 components retain 89%** of total variance, meeting the 85% target. The choice should also consider the needs of the application.

# PCA for Visualization

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# High-dimensional data in two dimensions

PCA can project high-dimensional observations onto **two principal components** for visualization.

For example, reducing 20 features to 2:

$$X \in \mathbb{R}^{n \times 20} \longrightarrow Z \in \mathbb{R}^{n \times 2}$$

We can then plot $PC_1$ **against** $PC_2$.

## What can a PCA plot reveal?

- Possible clusters or groupings.
- Unusual observations (**outliers**).
- Separation or overlap between groups.
- Broad patterns of variation.

**Important.** A 2D PCA plot may hide structure present in later components. Apparent separation in the plot does not automatically imply meaningful clusters.

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# PCA Before Supervised Learning

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# Useful but not always beneficial

PCA can be added to a **supervised learning pipeline** before training a predictive model.

## Potential benefits

- Reduce the number of input features.
- Summarize correlated or redundant information.
- Reduce noise in some datasets.
- Decrease computational cost.
- Sometimes improve distance-based methods.

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## Important limitation

PCA identifies directions of **high variance**, but it does **not use target values** $y$. A low-variance direction might still be highly predictive.

> **Evaluate the complete pipeline.** Fit scaling and PCA within each **training fold**, then compare downstream validation performance against a model **without PCA**.

# PCA Limitations

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# What PCA can and cannot capture

PCA is useful when strong directions of **linear variation** provide a compact representation of numerical data.

## Works well when

- Major patterns are approximately **linear**.
- High-variance directions are informative.
- Features are numerical or suitably encoded.
- Centering and scaling are appropriate.
- A compact representation is valuable.

## Important limitations

- Captures only **linear combinations**.
- Sensitive to **scaling and outliers**.
- Components may be difficult to interpret.
- Does not use target labels.
- May discard **low-variance but useful** information.

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💡 **Key idea.** PCA is a useful tool for compression, visualization, and preprocessing—but the best representation depends on **what information the application needs to preserve**.

# Part VI

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# Evaluating Unsupervised Learning

Without known target labels, evaluating unsupervised models requires more than a single score.

We examine **structure, stability, interpretation, and practical usefulness**.

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# Why Evaluation is Harder

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# No predefined correct answer

In **supervised learning**, predictions can be compared with known target values.

In **unsupervised learning**, there are usually no target labels defining the correct groups or representations.

## How can we evaluate the results?

- **Internal metrics:** Measure properties such as compactness and separation.
- **Stability:** Check whether patterns persist across repeated analyses.
- **Visualization:** Inspect group structure, overlap, and outliers.
- **Domain knowledge:** Assess whether the results make sense.
- **External information:** Compare with known categories if available.
- **Downstream usefulness:** Test whether the representation supports useful tasks.

💡 **Key idea.** Unsupervised evaluation asks whether the discovered patterns are **coherent, stable, and meaningful**, rather than simply whether predictions match known targets.

# Internal Clustering Metrics

FCIS — Private Use Only

# Useful scores, but not the whole story

**Internal metrics** evaluate clustering using the features and assigned groups, without requiring ground-truth labels.

| Metric | What it measures | Preferred value |
|---|---|---|
| **WCSS / inertia** | Squared distances to cluster centroids | Lower, for a fixed $K$ |
| **Silhouette score** | Cohesion within clusters and separation between clusters | Higher |
| **Davies–Bouldin index** | Cluster spread relative to separation | Lower |

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## Important limitations

- **WCSS** generally decreases as $K$ increases, so the smallest WCSS does not identify the best $K$.
- **Silhouette and Davies–Bouldin** favor particular geometric properties of clusters.
- A good numerical score does not guarantee a **meaningful real-world grouping**.

**Key caution.** Internal metrics describe mathematical properties of a clustering. Interpret them alongside **stability and domain knowledge**.

# Stability

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# Do the discovered groups persist?

**Cluster stability** examines whether similar groupings appear when the analysis is repeated under reasonable changes.

## What can we vary?

- **Initialization:** Start the algorithm from different positions.
- **Resampling:** Use different samples of the observations.
- **Small perturbations:** Introduce minor changes to the data.
- **Feature selection:** Check sensitivity to the included features.
- **Model settings:** Examine how group structure changes with parameters such as $K$.

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## Interpretation

- **Stable clusters:** Similar structures repeatedly emerge.
- **Unstable clusters:** Small changes produce substantially different groupings.

**Key lesson.** Stability increases confidence in a result, but **does not prove that the clusters are meaningful**. Highly unstable clusters deserve further investigation.

# Domain Validation

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# Does the clustering make sense?

After examining numerical metrics and stability, interpret the clusters in their **application context**.

## Questions to investigate

1. **Interpretability:** Can we describe what characterizes each group?
2. **Distinctiveness:** Which features separate the groups?
3. **Domain knowledge:** Do the patterns agree with what is already known?
4. **New insights:** Do the clusters reveal useful, unexpected structure?
5. **Group sizes:** Are any clusters unusually small or suspicious?
6. **Preprocessing:** Could scaling or irrelevant features explain the grouping?
7. **Usefulness:** Can the results support analysis or practical decisions?

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**Customer segmentation:** A group labeled *high-value customers* should actually show relevant spending and purchasing patterns. The label must be justified by examining the data, not merely by the cluster number.

💡 **Key idea.** Clustering becomes valuable when the discovered groups are **understandable, credible, and useful for the intended purpose**.

# Part VII

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# Worked Mini-Cases

**Unsupervised learning** becomes useful when discovered patterns are **interpreted, validated, and connected to meaningful decisions**.

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# Mini-Case: Student Engagement Clustering

Public - For Course Use Only

# Understanding engagement patterns

A university wants to identify **patterns of student engagement** without predefined student categories.

## Available features

| Feature | What it describes |
|---|---|
| LMS logins per week | Online activity |
| Lab submissions | Practical engagement |
| Attendance percentage | Classroom participation |
| Forum posts | Online interaction |
| Average quiz score | Early academic performance |

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## Goal

Discover **student engagement profiles** that may help the university design appropriate support.

> **Task: Clustering.** No target labels are provided. The algorithm discovers groups based on similarities in the selected features.

# Mini-Case: Preprocessing Choices

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# Prepare features before clustering

Before clustering students, several decisions matter:

- **Scaling:** Put numerical features on comparable scales.
- **Missing values:** Check and handle incomplete records.
- **Outliers:** Examine unusually high or low activity.
- **Feature relevance:** Decide whether quiz scores belong in the analysis.
- **Timing:** Use information available at the intended decision time.
- **Documentation:** Record how features were defined and processed.

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## Important question

> If the goal is **early student support**, should final course grades be included?

> **Avoid future information.** Final grades would not be available when early support decisions are made. Including them could produce misleading profiles that cannot be reproduced at that time.

# Mini-Case: Interpreting Clusters

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# From numerical groups to student profiles

Suppose clustering reveals four **illustrative engagement patterns**:

| Cluster | Observed pattern | Possible interpretation |
|---|---|---|
| **C1** | High attendance, submissions, and logins | Strong overall engagement |
| **C2** | High logins, low attendance | Online-focused engagement |
| **C3** | Low observed activity across features | Limited recorded engagement |
| **C4** | High attendance, low quiz scores | Attending but struggling academically |

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## How might the university respond?

Investigate whether different groups would benefit from **academic guidance, flexible participation options, or targeted learning support**.

> **Important.** These profiles are **interpretations, not fixed student identities**. Validate them and consider individual circumstances before making supportive, non-punitive decisions.

# Mini-Case: PCA for Visualization

FCIS — Course Use Only

# Reduce five engagement features to two dimensions

Suppose PCA produces these **illustrative explained-variance ratios**:

| Component | Explained variance |
|---|---|
| $PC_1$ | 58% |
| $PC_2$ | 21% |
| **Total retained** | **79%** |

A plot of $PC_1$ **versus** $PC_2$ therefore represents 79% of the dataset's total variance.

## Possible interpretation

- $PC_1$**:** May reflect overall engagement.
- $PC_2$**:** May distinguish online from classroom participation.

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💡 **Key idea.** PCA can help visualize student patterns in 2D, but **component meanings require examining loadings**. The remaining **21% of variance** is not shown.

# Practice & Consolidation

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# Apply the concepts

Can you identify an **unsupervised task**, perform a **K-Means update**, choose $K$, interpret **PCA**, and recognize misleading conclusions?

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# Activity: Identify the Task

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# Supervised or unsupervised?

For each scenario, identify the learning type and explain your choice.

| Scenario | Learning type | Why? |
|---|---|---|
| Predict exam scores from study data | | |
| Group customers by purchasing behavior | | |
| Reduce 50 features to 2 for visualization | | |
| Predict whether a transaction is fraudulent | | |
| Discover document topics from text | | |

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# Supervised or unsupervised?

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| --- | --- | --- |
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| Group customers by purchasing behavior | | |
| Reduce 50 features to 2 for visualization | | |
| Predict whether a transaction is fraudulent | | |
| Discover document topics from text | | |

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**Solution**

- **Supervised:** Exam-score prediction and fraud classification use known target values.
- **Unsupervised:** Customer clustering, PCA visualization, and discovering document topics do not require predefined target labels.

# Activity: One K-Means Iteration

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# Assign points and update centroids

Consider four observations:

$$A = (0, 0), \quad B = (0, 2), \quad C = (5, 5), \quad D = (6, 5)$$

Initial centroids:

$$\mu_1 = (0, 0), \qquad \mu_2 = (5, 5)$$

## Tasks

1. Assign each point to its **nearest centroid**.
2. Calculate the **updated centroids**.
3. Check whether another assignment step changes the clusters.

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# Assign points and update centroids

Consider four observations:

$$A = (0, 0), \quad B = (0, 2), \quad C = (5, 5), \quad D = (6, 5)$$

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## Tasks

1. Assign each point to its **nearest centroid**.
2. Calculate the **updated centroids**.
3. Check whether another assignment step changes the clusters.

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**Solution**

- **Assignments:** $C_1 = \{A, B\}$ and $C_2 = \{C, D\}$.
- **Updated centroids:** $\mu_1 = (0, 1)$ and $\mu_2 = (5.5, 5)$.
- **Convergence:** Reassigning the points does not change the groups, so the algorithm has converged.

# Activity: Choose $K$

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# Interpret the WCSS values

Suppose K-Means produces these **illustrative results**:

| $K$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **WCSS** | 1200 | 620 | 330 | 290 | 260 | 245 |

## Questions

1. Where is a plausible **elbow**?
2. Why does WCSS generally decrease as $K$ increases?
3. Why is WCSS alone insufficient for choosing $K$?

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# Interpret the WCSS values

Suppose K-Means produces these **illustrative results**:

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## Questions

1. Where is a plausible **elbow**?
2. Why does WCSS generally decrease as $K$ increases?
3. Why is WCSS alone insufficient for choosing $K$?

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**Solution:** A plausible elbow is $K = 3$: WCSS decreases sharply before this point and more slowly afterward. More centroids allow a lower (or equal) optimal WCSS, but **stability, interpretability, and usefulness** must also guide the choice.

# Activity: PCA Explained Variance

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# How many components should we retain?

Suppose PCA produces the following explained-variance ratios:

$$0.45, \quad 0.25, \quad 0.15, \quad 0.10, \quad 0.05$$

## Questions

1. What percentage of variance is explained by the **first two components**?
2. How many components are needed to preserve **at least 80%** of the variance?
3. Why might retaining 80% of variance be insufficient for a **supervised prediction task**?

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# How many components should we retain?

Suppose PCA produces the following explained-variance ratios:

$$0.45, \quad 0.25, \quad 0.15, \quad 0.10, \quad 0.05$$

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3. Why might retaining 80% of variance be insufficient for a **supervised prediction task**?

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**Solution**

1. First two components: $0.45 + 0.25 = \boxed{70\%}$.
2. First three components: $0.45 + 0.25 + 0.15 = \boxed{85\%}$, meeting the 80% target.
3. PCA ignores target labels. A component with **low variance** may still contain important predictive information.

# Common Mistakes

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# Unsupervised learning pitfalls

| Mistake | Why it matters |
|---|---|
| **Treating clusters as true labels** | Groups are algorithmic outputs, not established categories |
| **Clustering without appropriate scaling** | Large-scale features may dominate distances |
| **Choosing $K$ mechanically** | Good numerical scores may not mean useful clusters |
| **Ignoring initialization** | K-Means can produce different solutions |
| **Naming clusters without inspection** | Interpretations may not match the feature patterns |
| **Applying PCA without considering scale** | High-variance features may dominate components |
| **Equating variance with predictive value** | PCA does not use target labels |
| **Using accuracy without ground truth** | There is no known correct label to compare against |

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💡 **Final lesson.** Unsupervised learning requires more than running an algorithm: **prepare data carefully, evaluate stability and structure, and interpret results in context**.

# Mini-Quiz

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# Check your understanding

1. What is the main difference between supervised and unsupervised learning?
2. What does a K-Means centroid represent?
3. What are the two main steps repeated by K-Means?
4. Why does scaling matter for K-Means?
5. Why can different K-Means initializations lead to different results?
6. What does the elbow method try to identify?
7. What does PCA preserve?
8. Why can PCA components be difficult to interpret?

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# Check your understanding

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4. Why does scaling matter for K-Means?
5. Why can different K-Means initializations lead to different results?
6. What does the elbow method try to identify?
7. What does PCA preserve?
8. Why can PCA components be difficult to interpret?

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**Suggested answers:** supervised learning uses labels; a centroid is the cluster mean; assignment and update; distances depend on scale; local optima; diminishing improvement in WCSS; variance; components are linear combinations of original features.

# Lecture 8 Summary

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# Key ideas to retain

## Clustering

- **Unsupervised learning** discovers structure in $X$ without target labels.
- **Clustering** groups observations based on similarity.
- **K-Means** alternates between nearest-centroid assignment and centroid updates.
- **Scaling and initialization** influence clustering results.
- Choose $K$ using metrics, stability, and domain usefulness.
- **Hierarchical clustering** builds nested groups represented by a dendrogram.

## Dimensionality Reduction

- Reduces the number of features while preserving important structure.
- **PCA** creates new features as linear combinations of original variables.
- Principal components capture directions of **maximum variance**.
- **Explained variance** helps determine how many components to retain.
- PCA supports **visualization, compression, and preprocessing**.
- Discovered clusters and components require **careful interpretation**.

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# End-of-Lecture Check

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# Five questions to test your understanding

1. Why is **clustering** considered unsupervised learning?
2. What does **K-Means minimize**, and how do its assignment and update steps work?
3. Why should the choice of $K$ consider more than WCSS alone?
4. How does **PCA** differ from feature selection?
5. Why might a clustering or PCA result look convincing but still be **misleading or unhelpful**?

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💡 **Course connection.** Machine learning is not limited to predicting known targets. **Unsupervised learning** discovers patterns without labels—but trustworthy results still require careful preprocessing, evaluation, and interpretation.

# Suggested Reading

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# Primary and supporting references

## Primary reference

**G. James, D. Witten, T. Hastie, and R. Tibshirani,** *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.

- Review unsupervised learning, clustering, K-Means, hierarchical clustering, and PCA.

## Supporting references

**A. Géron,** *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly, 2022.

- Review clustering, preprocessing, dimensionality reduction, and practical Scikit-Learn examples.

**K. P. Murphy,** *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.

- Use selected sections for deeper views of latent structure, clustering, and probabilistic interpretation.

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💡 When reading, focus on how unsupervised methods produce structure and how that structure should be interpreted and validated.